

Integrating Cyber Threat Intelligence into Threat Modeling for Autonomous Ships Using PASTA and MISP

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Abstract—Autonomous maritime systems rely on the complex interplay of operational technology (OT) and information technology (IT), making them increasingly vulnerable to cyber threats. This paper presents an integrated approach to autonomous ship cybersecurity by combining the Process for Attack Simulation and Threat Analysis (PASTA) threat modeling framework with Malware Information Sharing Platform (MISP) for cyber threat intelligence (CTI). The autonomous ship architecture is divided into three security layers, which include OT and IT assets, decision-making entities, and decision inputs. CTI is systematically integrated into dynamic threat models by mapping maritime-specific attack models to MITRE ATT&CK and MITRE ATLAS, enabling a structured and adaptive threat representation. The results demonstrate that integrating MISP with PASTA can improve situational awareness, facilitate real-time updates of threat models, and improve cybersecurity in autonomous maritime operations.

Index Terms—Autonomous Ships, Maritime Cybersecurity, PASTA, Cyber Threat Intelligence, MISP

I. INTRODUCTION

The development of autonomous ship systems is evolving as maritime industries seek to improve operational efficiency and safety by reducing human error. These Maritime Autonomous Surface Ships (MASS) are complex cyber-physical systems (CPS) that integrate advanced sensors, artificial intelligence (AI) based decision-making and networked communications to navigate with minimal human intervention [1]. Major projects such as MUNIN and Yara Birkeland exemplify progress in MASS development and point to a future where ships can operate autonomously in the oceans. But alongside these advances, the cybersecurity of MASS has emerged as a critical concern. Increased reliance on digital technologies and connectivity expands the attack surface, leaving essential ship systems vulnerable to potential exploitation by malicious actors. Despite the high risks, securing MASS is challenging due

to their complex system design and the constant advancement of cyber threats.

Cyber threats are a major challenge for MASS because a successful attack can compromise both the digital control and physical security of the ship. Unlike traditional manned ships, MASS as defined by the International Maritime Organization (IMO)¹ are particularly vulnerable to cyberattacks, as there is no crew to detect anomalies or take manual corrective action. Threats range from GPS spoofing and sensor manipulation to malware infiltrating control systems or AI decision algorithms compromised by attacker inputs. Jahan et al. [2] note that cybersecurity research into autonomous systems is still in its infancy, and advanced threat modeling techniques are needed to address emerging threat vectors. In the maritime domain, several frameworks for cyber risk assessment have been proposed, but there is a lack of integrated solutions that can adapt to new threats in real-time. Existing maritime cybersecurity frameworks primarily rely on static risk assessments, which fail to account for emerging cyber threats. These methods may not incorporate up-to-date cyber threat intelligence (CTI) streams, causing a gap in situational awareness as new vulnerabilities and attacker tactics emerge.

In this paper, we propose to integrate the Process for Attack Simulation and Threat Analysis (PASTA) threat modeling framework with the Malware Information Sharing Platform (MISP) CTI system to create an adaptive cybersecurity methodology for MASS. The goal of this integration is to leverage the systematic approach to identify, model, and analyze threats provided by PASTA, while using MISP to inject real-time CTI into the modeling process. In doing so, our goal

¹International Maritime Organization (IMO), *Autonomous Shipping*, <https://www.imo.org/en/MediaCentre/HotTopics/Pages/Autonomous-shipping.aspx>

is to continuously update the cyber threats considered in the model, increasing its relevance and effectiveness.

While PASTA provides a structured and systematic approach to threat modeling, it remains a static framework that does not adapt to emerging threats. This means that new vulnerabilities, zero-day exploits, and adversarial AI tactics may not be detected if they were not considered during the initial modeling phase. To address this, we integrate MISP into PASTA to enable real-time threat intelligence updates.

II. BACKGROUND

MASS represent a breakthrough, operating with various degrees of autonomy, from partial automation to full autonomy [3]. These ships use a range of sensors and data analytics for precise navigation and decision-making, supported by advanced communication systems for seamless data exchange with other ships and operations ashore. Integrated control systems at the core of a MASS synthesize sensor data to navigate and execute operational commands in compliance with maritime regulations. The IMO defines several levels of autonomy, from vessels that enhance crew capabilities with autonomous technologies to MASS capable of operating without human intervention. Given that Level 4 has not yet been reached, the research will advance efforts towards fully autonomous MASS [4].

MASS relies on AI and machine learning (ML) models for tasks such as collision avoidance, path planning and anomaly detection. These AI components introduce new types of vulnerabilities not found in traditional ships. For example, an attacker could exploit an AI model through adversarial ML techniques by feeding it intentionally crafted inputs to deceive the model's pattern recognition [5]. This suggests that AI algorithms can be manipulated in ways that standard cybersecurity frameworks cannot capture. In response to such threats, MITRE² has created the ATLAS³ knowledge base as a globally accessible source for adversary tactics and techniques against AI-enabled systems based on real-world attack observations. Since MASS heavily rely on AI-based decision-making, the attack techniques outlined in MITRE ATLAS provide valuable insights into AI-specific threats that standard cybersecurity models may overlook. By referring to frameworks such as MITRE ATT&CK for general cyber threats and adversary tactics and techniques, and MITRE ATLAS for AI-specific threats, analysts can systematically identify potential attack techniques that a MASS might face.

PASTA's⁴ structured approach is well suited for complex systems such as MASS. By emphasizing system decomposition and comprehensive threat identification, PASTA enables analysts to consider the full breadth of the system and various threat possibilities. Previous applications of PASTA in domains such as web applications and IoT systems have demonstrated its effectiveness in uncovering non-obvious at-

tack vectors and aligning security efforts with the most critical assets [6].

The outcome of a PASTA exercise is a threat model and a set of prioritized security measures, showing how a threat actor could mount an end-to-end attack on the system and what the consequences would be. By focusing on the impacts, PASTA helps to ensure that the mitigations proposed are proportionate to the risk.

The selection of PASTA is motivated by several advantages. PASTA follows a structured, seven-step approach, making it particularly well-suited for complex MASS. Unlike STRIDE and ATT&CK, which focus primarily on attacker tactics and system vulnerabilities, PASTA aligns with operational risk management, making it more applicable to maritime cybersecurity. Furthermore, PASTA's system decomposition step proved advantageous in this research, as it facilitated the decomposition of the complex MASS into functional components, making it easier to identify vulnerabilities and model attack scenarios. Moreover, PASTA's multi-stage threat modeling approach differentiates it from other methodologies by systematically transitioning from threat identification to risk assessment and mitigation, ensuring that cyber risks are evaluated within the context of operational impact.

The MISP⁵ is a widely used open source platform for the collection, storage and sharing of CTI between trusted organizations. Originally developed to facilitate the exchange of information about malware and cyberattacks, MISP has evolved into a community-driven CTI hub that supports structured data on Indicators of Compromise (IoCs), threat actor profiles, attack techniques and security incidents. MISP provides a database where users can input and retrieve IoCs such as malicious IP addresses, domain names, malware file hashes, phishing email attributes and vulnerability identifiers. MISP enables organizations to share information in a timely manner about threats under a configurable sharing policy. As a result, when one organization uploads details of a new attack or IoC, others can quickly learn from it and update their defenses.

III. RELATED WORK

In this section, we review the most relevant literature on threat modeling approaches applied to maritime systems and the role of CTI in cybersecurity.

Maritime cybersecurity models primarily addressed conventional ships before adapting to MASS. The Maritime Cyber Risk Assessment (MaCRA) model [7] provided a structured evaluation of cyber risks, later extended by Tam and Jones using a multi-criteria decision-making (MCDM) framework for MASS [8]. Similarly, Kavallieratos et al. [9] emphasized cybersecurity controls in digitized ships, while Bolbot et al. [10] introduced CYRA-MS, a quantitative risk assessment method for maritime cyber threats. However, these models rely on static assessments without integrating real-time CTI.

²MITRE ATT&CK: <https://attack.mitre.org/>

³MITRE ATLAS: <https://atlas.mitre.org/>

⁴PASTA: <https://versprite.com/blog/pasta-threat-modeling-solution/>

⁵MISP: <https://www.misp-project.org/>

A more detailed approach emerged with the adaptation of the MITRE ATT&CK framework to maritime cybersecurity. Oruc et al. [11] mapped attack tactics to a ship's navigation system, while Amro et al. [12] combined ATT&CK with Failure Modes, Effects and Criticality Analysis (FMECA) to assess risks in autonomous cruise ships. While these studies enhance technology-specific threat analysis, they still have notable limitations as identified in the systematic literature review by Erbas et al. [13]. Existing maritime threat modeling methodologies lack standardized frameworks that can adapt dynamically, while relying on static threat models that do not account for real-time updates from CTI platforms.

While MISP has been widely recognized as a powerful CTI-sharing platform in the broader cybersecurity community, its integration with maritime cybersecurity threat modeling remains largely unexplored. Research on MISP applications in cybersecurity has demonstrated its ability to improve situational awareness, accelerate incident response, and enhance security decision-making. Stojkovski et al. [14] evaluated MISP from a user-experience perspective, reinforcing its role in streamlining CTI sharing across organizations. Similarly, Alzahrani and Lee [15] explored the integration of MISP with Security Information and Event Management (SIEM) systems, demonstrating how it can aggregate CTI from multiple sources including honeypot and Open Source Intelligence (OSINT).

Despite these advancements, no existing research has explored how MISP can be integrated with maritime threat modeling frameworks. This paper addresses this gap by proposing a PASTA-MISP integration, allowing for a real-time, CTI-driven maritime cybersecurity framework. By linking PASTA's structured threat modeling approach with MISP's dynamic CTI capabilities, we enable dynamic updating of threat models based on attack trends and real-time CTI feeds. This improves situational awareness and adaptability, ensuring that cybersecurity defenses evolve in tandem with new threats.

IV. METHODOLOGY

PASTA provides a structured, and risk-centric methodology designed for comprehensive threat modeling across complex systems. PASTA consists of seven steps: defining objectives, decomposing the system architecture, identifying potential threat agents, cataloging threats to critical system functions, assessing vulnerabilities, modeling attack paths, and proposing mitigation strategies. PASTA is well-suited for MASS, where IT, OT, AI-driven decision-making, and communication systems interact across multiple layers. The complexity of MASS requires a systematic modeling approach, which PASTA provides. To enhance the dynamic quality of threat modeling, we integrate PASTA with CTI using MISP, enriching each modeling stage with up-to-date intelligence about vulnerabilities, attacker behaviors, and incident data.

The objectives are to apply PASTA to various layers of a MASS architecture, identify potential vulnerabilities and attack paths, and incorporate MISP to enrich each stage of threat modeling with up-to-date information on threats, enabling dynamic re-assessment of risk and the development of efficient

mitigation strategies and security controls informed by the latest known threat models. We identify a range of threats and attack vectors for each layer of the MASS architecture, from low-level sensor spoofing attacks to high-level decision algorithm manipulations.

We do not investigate specific configurations of any particular ship's hardware or software, but instead develop a generalized layered model representing MASS from the literature and industry projects. With this abstraction, our findings and methodology remain applicable across different MASS designs and platforms.

The core of our methodology involves applying the structured threat modeling approach of PASTA within the MISP platform to represent cybersecurity threats systematically. To complement the structured modeling provided by PASTA, we integrate CTI using the MISP. The combination of PASTA and MISP is realized by mapping the outputs of each PASTA modeling layer into structured CTI events within MISP. For each system layer modeled in PASTA, a corresponding MISP event is created to capture the associated the attack types, vulnerabilities, and affected components. These events are enriched with CTI objects, such as IoCs, threat actor profiles, and attack techniques, all linked using standardized taxonomies including MITRE ATT&CK and MITRE ATLAS. This mapping enables validation of modeled threats through CTI and facilitates continuous updating of the threat model as new data becomes available in MISP. When new CTI data arrives in MISP, such as an updated attack technique or vulnerability, it is linked to the relevant system components in the threat model via standardized taxonomies, resulting in dynamic updates to threat paths, influencing risk assessment and mitigation strategies.

First, the MASS architecture is decomposed into three distinctive layers as shown in Table I to facilitate precise threat identification. Layer 1 concerns the primary OT and IT systems, including navigation and communication systems. Layer 2 concerns the autonomous decision-making and ML models. Layer 3 concerns the inputs for decision-making such as collision avoidance and data integration.

TABLE I
MISP THREAT INTELLIGENCE DATA FOR AUTONOMOUS SHIPS

Event ID	# Attributes	Info
3794	107	PASTA Threat Modeling for MASS
3795	51	PASTA - Layer 1: Primary Function Categories
3796	46	PASTA - Layer 2: Decision-Making and Autonomy
3797	28	PASTA - Layer 3: Inputs for Decision-Making

We follow PASTA's seven steps to analyze the security of a MASS. We begin by defining the security objectives and scope of analysis for the MASS IT and OT components. The primary objectives include ensuring safe navigation, maintaining operational continuity, and protecting communication integrity. We identify key assets whose compromise would be unacceptable:

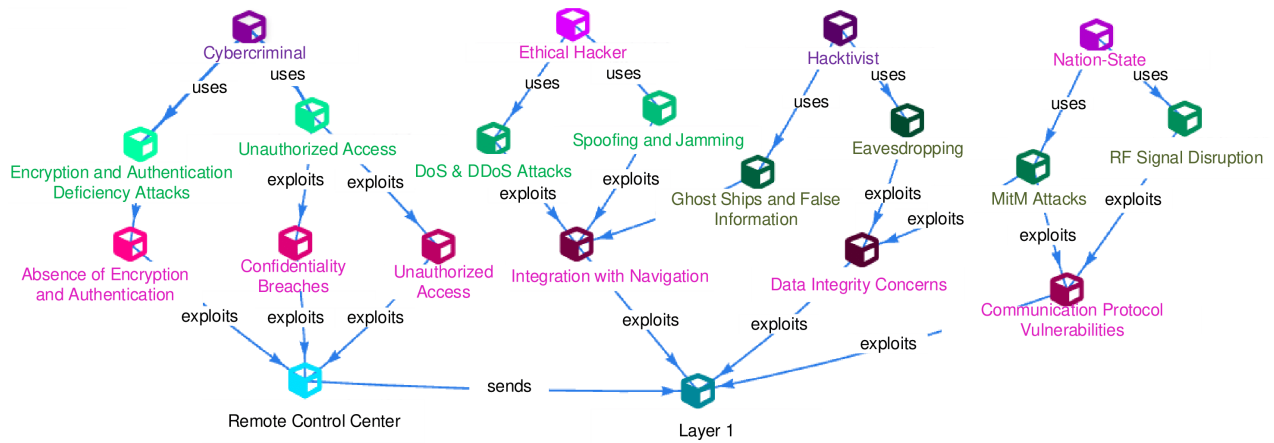


Fig. 1. Attack Model for Layer 1: Primary Functional Categories. This figure has been derived from the original MISP graphical data to present key elements in a more compact and comprehensible manner.

navigation accuracy, sensor data integrity, propulsion control, collision avoidance logic, and communication channels to shore. By clarifying these objectives, we set the focus for subsequent threat modeling steps on attack models that could undermine these essential aspects. We delineate the MASS architecture and break it down into components and subsystems, leveraging the layered model. Data flow diagrams illustrate the interactions among key functional components, including navigation, communication, propulsion, sensor analysis, and control systems. These interconnected elements support MASS operations and decision-making processes. Inputs from onboard systems, external data sources, and remote communication contribute to autonomous navigation and situational awareness. While we examine cybersecurity risks across all layers, our primary focus is on threats affecting autonomy, AI-driven decision-making, and data integrity.

Each layer is modeled individually, with threat levels categorized as shown in Fig. 2 with Event ID, identifying specific vulnerabilities, threats, and affected components, thus enabling precise analysis and contextualization of cyber risks. Following the decomposition, PASTA threat modeling methodology is employed systematically assessing threats, vulnerabilities, and attack paths through layers. Attack models are identified by tracing threat actors, their utilized attack types, and exploited vulnerabilities as demonstrated in Figs. 1, 3, and 5.

Event ID	3795
UUID	10194cec-0f6a-44b6-b826-1277ee588cd3
Threat Level	Medium

Fig. 2. Threat Level of Layer 1 in MISP

To enhance the relevance of the threat model, MISP is incorporated into the process by structuring CTI data into three MISP events, each corresponding to a system layer (as shown in Table I). Each threat, vulnerability, and attack technique identified through PASTA is mapped to structured CTI in MISP using standardized taxonomies such as MITRE

ATT&CK and MITRE ATLAS. This ensures that our threat modeling process is continuously updated with real-world IoCs and observed attack behaviors. Additionally, MISP's correlation capabilities are leveraged to validate the attack paths developed through PASTA, enabling cross-referencing with existing maritime cybersecurity incidents.

The standardized use of MITRE ATT&CK and ATLAS within MISP further enriches the actionable intelligence data, supporting precise threat identification, efficient analysis, and proactive response. Using the proposed methodology, we obtained a detailed analysis of threats and vulnerabilities for each layer of the MASS, along with a structured mapping of how MISP integration enhances threat modeling by dynamically incorporating CTI.

V. CASE STUDY: THREAT MODELING AND CTI FOR AI-BASED AUTONOMOUS SHIP DECISION-MAKING

The methodology steps were applied to a replicated setup based on a copy of an operational MISP instance utilized in maritime cybersecurity by the Turku University of Applied Sciences. As shown in Fig. 4, the replicated MISP instance retained structured event entries, including unique UUIDs and descriptions, verifying that the testing environment was consistent with an operational setting. This approach allowed us to test the integration of PASTA with MISP under conditions that closely resembled a real-life MISP environment, while avoiding interference with active security operations. The replicated system contained CTI data, enabling structured testing of the threat modeling integration and its validation in a realistic test environment, ensuring that the approach remains applicable to real-world maritime cybersecurity scenarios.

The integration of PASTA with MISP demonstrated enhanced visibility into the cybersecurity of MASS. While PASTA alone provides a structured methodology for modeling cyber threats, it lacks the ability to validate modeled threats against real-world attack intelligence, prioritize emerging threats based on observed attack frequency, and dynamically update threat models as adversarial tactics evolve.

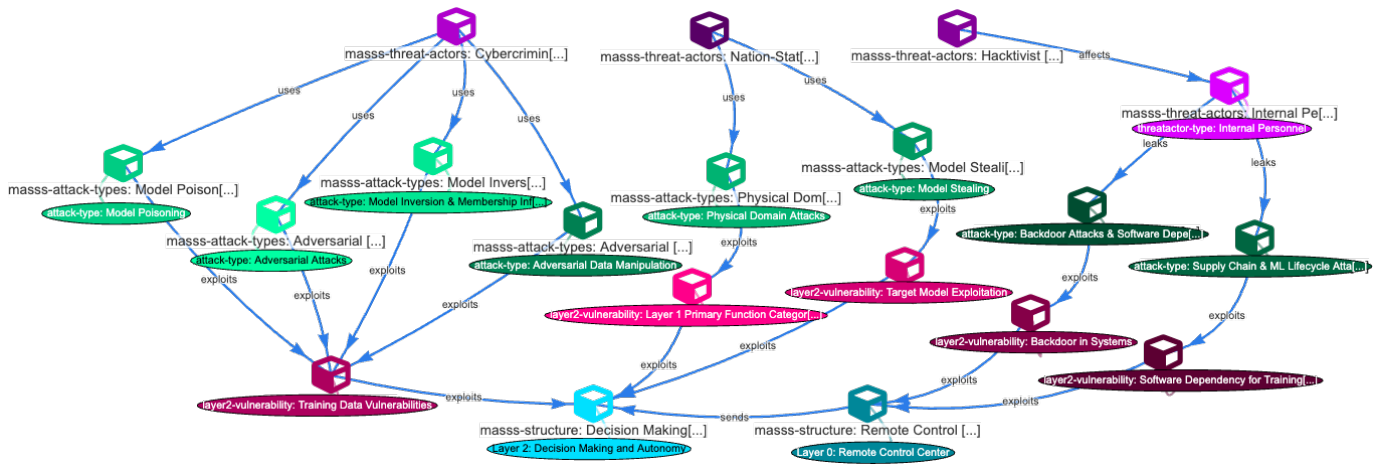


Fig. 3. Attack Model for Layer 2: Decision Making and Autonomy.

Local organisations		
Known remote organisations		
All organisations		
ID	Name	UUID
61	6be	6be

Fig. 4. MISP Event Entry Showing Organization

This is where MISP plays an important role. The structured representation provided by MISP allows us to cross-reference PASTA threat models with cyber intelligence data, ensuring that our findings align with actual attack campaigns observed in maritime cybersecurity. Additionally, MISP enhances situational awareness by mapping attack types, vulnerabilities, and affected systems to threat actor behaviors, allowing for risk-based prioritization of mitigation efforts.

To highlight the benefits of integrating MISP with PASTA, we analyze a practical example of cyber threat detection in autonomous ship systems across three security layers. The results were organized by system layer and by the impact of threat intelligence on our threat model, demonstrating how the PASTA and MISP integration transforms static attack modeling into a dynamic and intelligence-driven approach.

Our analysis for Layer 1 identified attack vectors targeting the ship's sensor and control subsystems (see Fig. 1), with threat actors exploiting vulnerabilities in navigational devices, such as unauthorized access and encryption deficiencies, enabling false data transmission that lead to misclassifications and compromised situational awareness. These vulnerabilities can directly impact Layer 2 decision-making, ultimately affecting navigation and collision avoidance in Layer 3.

The threats in Layer 2 revolved around adversarial attacks that undermined the logic and integrity of autonomous decision-making algorithms (see Fig. 3). These attacks exploit training data vulnerabilities, model poisoning, and model inversion techniques, allowing threat actors to manipulate AI behavior and degrade decision reliability.

In Layer 3, the Autonomous Navigation Decision Layer,

compromised inputs from Layer 1 and manipulated decision logic from Layer 2 can lead to incorrect course adjustments, collision risks, and navigational errors (see Fig. 5). This layer integrates sensor data, decision-making outputs, and collision avoidance mechanisms, making it critical for ensuring safe and reliable autonomous operations.

After identifying and modeling cyber threats across the three layers using the PASTA methodology, we transitioned to integrating these findings into MISP for validation and intelligence-sharing. The structured attack models from PASTA served as a foundational representation of vulnerabilities, attack types, and affected system components. However, to further enhance the accuracy and applicability of these models, MISP was used to correlate modeled threats with CTI.

Based on the structured attack models developed using PASTA, we integrated these findings into MISP by creating separate events for each architectural layer. Each MISP event corresponds to an attack scenario identified in PASTA, allowing for a structured representation of threats, vulnerabilities, and attack types. This process enabled a direct correlation between attack models and CTI, as shown in Table II. Each MISP event captured vulnerabilities, attack types, and system components, all linked via detailed objects (see Fig. 6) and tags using standardized taxonomies (see Fig. 7).

Each threat model developed using PASTA was transformed into a structured MISP event, where system vulnerabilities, attack types, and affected components were systematically mapped using standardized taxonomies such as MITRE ATT&CK and ATLAS. Additionally, each attack path modeled in PASTA was cross-referenced with MISP's structured IoCs and threat actor profiles, further strengthening the accuracy of attack modeling. Table II presents the mapping between PASTA threat model outputs and their corresponding representation in MISP.

MITRE ATT&CK and MITRE ATLAS standards are used within the MISP activities to ensure interoperability with the wider cybersecurity intelligence community. At this stage, the

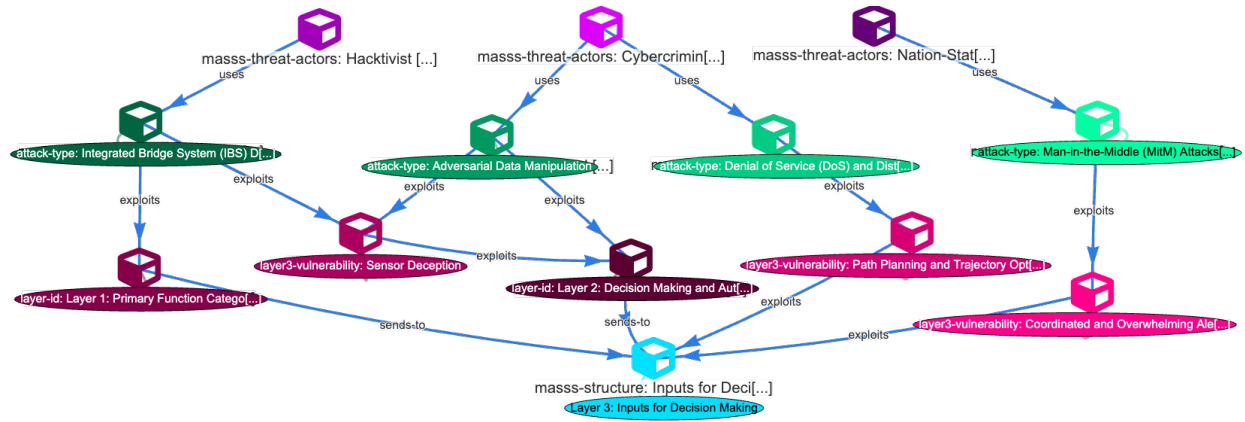


Fig. 5. Attack Model for Layer 3: Inputs for Decision Making.

Objects

- masss-threat-actors: Ethical Hacker
- masss-threat-actors: Nation-State
- masss-threat-actors: Hacktivist
- masss-threat-actors: Cybercriminal
- masss-attack-types: Eavesdropping
- masss-attack-types: Ghost Ships and False Information
- masss-attack-types: Man-in-the-Middle (MitM) Attacks
- masss-attack-types: RF Signal Disruption
- masss-attack-types: Denial of Service (DoS) and Distributed Denial of S...
- masss-attack-types: Spoofing and Jamming
- masss-attack-types: False Data Transmission and Identity
- masss-attack-types: Unauthorized Access and Emergency Protocol Ta...
- masss-attack-types: Encryption and Authentication Deficiency Attacks
- masss-vulnerabilities: Data Integrity Concerns
- masss-vulnerabilities: Integration with Navigation
- masss-vulnerabilities: Communication Protocol Vulnerabilities
- masss-vulnerabilities: Unauthorized Access
- masss-vulnerabilities: Confidentiality Breaches
- masss-vulnerabilities: Absence of Encryption and Authentication
- masss-structure: Navigation System
- masss-structure: Remote Control Center

Fig. 6. MISP Objects Representing Threat Actors, Attack Types, Vulnerabilities, and Structures.

MISP integration involved associating threats with vulnerabilities, and we started to build attack paths, synthesizing information from the PASTA model on how attacks could progress through the system. We validated each step in the attack models by cross-referencing MISP threat intelligence data with known attack scenarios from literature. The attack types, vulnerabilities, and structures used were matched and validated against each object based on studies in the literature. As a result, one or more representative attack models were created at each layer, resulting in a layered attack matrix.

For example, in Fig. 8, the validation process for Layer 2 of the attack model involved verifying the Model Poisoning attack type. This layer specifically addressed adversarial AI techniques where malicious modifications can be introduced

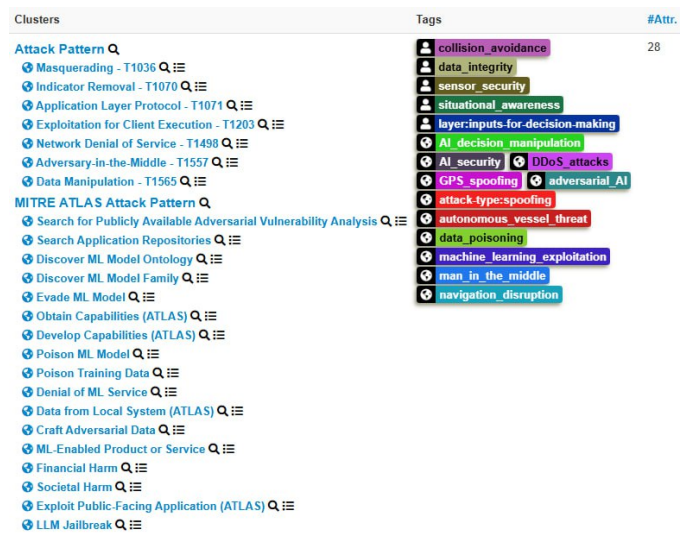


Fig. 7. MISP Clusters and Tags for Layer 3

TABLE II
MAPPING OF PASTA THREAT MODEL ELEMENTS TO MISP DATA

PASTA Output	How It Was Represented in MISP
Identified threats and vulnerabilities	Structured as MISP attributes with linked objects
Attack types and paths	Defined within MISP as correlated attack scenarios
System layers (Layer 1, Layer 2, Layer 3)	Created as separate MISP events for structured representation
Threat actor associations	Mapped using MITRE ATT&CK and ATLAS frameworks

into the training phase, leading to long-term model degradation. To validate this, we cross-referenced structured attack data from MISP with documented adversarial AI techniques.

The Model Poisoning attack type was mapped within MISP to the broader category of Training Data Manipulation attacks. This correlation was validated against documented adversarial AI techniques, such as those observed in Walter et al. [5],

Category	Type	Value
External analysis	other	Walter, M., Barrett, A., Walker, D., & Tam, K. (2023). Adversarial AI Testcases for Maritime Autonomous Systems. <i>AI, Computer Science and Robotics Technology</i> , 6. https://doi.org/10.5772/acrt.15
Object name:	mass-attack-types []	attack-type :: text
References:	2 []	Model Poisoning
Referenced by:	1 []	
		Hide 3 Attributes

Fig. 8. Model Poisoning Attack Using MISP Threat Intelligence Data

which demonstrated how adversarial test cases in maritime autonomous systems can degrade classification accuracy. By linking our modeled attack paths with CTI, we can ensure that our findings align with known adversarial AI strategies targeting autonomous decision-making systems. The attack model for Data Poisoning resulted in a structured representation of the relationships between threat actors, attack types, vulnerabilities, and their impact on the Layer 2 decision-making model (see Fig. 9). This attack can exploit training data vulnerabilities at Layer 2, impacting decision-making and autonomy in AI-driven systems.

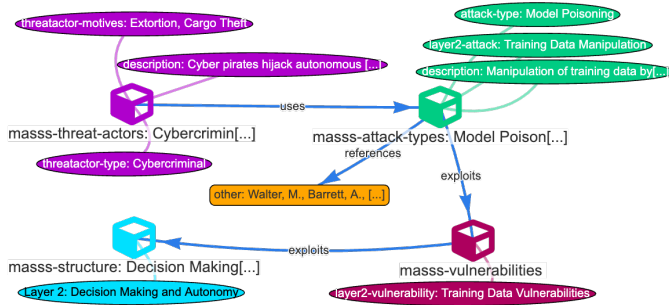


Fig. 9. Attack Model for Data Poisoning

Each threat model developed using PASTA was validated through a structured mapping to CTI in MISP. The data poisoning model (see Fig. 9) was cross-referenced with known adversarial AI strategies research in Walter et al. [5] and linked with MISP attributes such as attack types, vulnerabilities, and system structures. This method of structured validation enables alignment between modeled threats and actual observed attack behavior.

VI. CONCLUSION

This paper presented an integrated threat modeling approach for autonomous ships by combining PASTA with MISP CTI. Our findings demonstrate that securing inter-layer communications and ensuring data integrity are critical to protecting autonomous ship systems. The integration of MITRE ATT&CK and ATLAS strengthens the threat modeling process by providing structured representations of attack techniques. The PASTA and MISP approach identifies emerging threats, including adversarial AI manipulations, supply chain vulnerabilities, and CPS exploits, which traditional static models may overlook. This results in a cybersecurity framework that continuously evolves with real-world attack intelligence.

This study also presents certain limitations. Integrating real-time threat intelligence through MISP into the PASTA threat modeling framework increases the volume and variability of input data, which can create challenges in maintaining the accuracy, consistency, and relevance of threat models over time. The inclusion of dynamic intelligence requires continuous validation to avoid misinterpretation of threat indicators, particularly when threat data contains incomplete or low-confidence information. Although our methodology improves the alignment of threat models with real-world attack behaviors, future steps involve testing in an operational environment. In this study, we also considered both manual and automated ingestion of CTI into MISP to enable flexible and adaptive threat model updates. We will focus on automating CTI ingestion pipelines within MISP, expanding maritime-specific system and attack taxonomies, and developing real-time countermeasures against newly identified cyber threats.

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